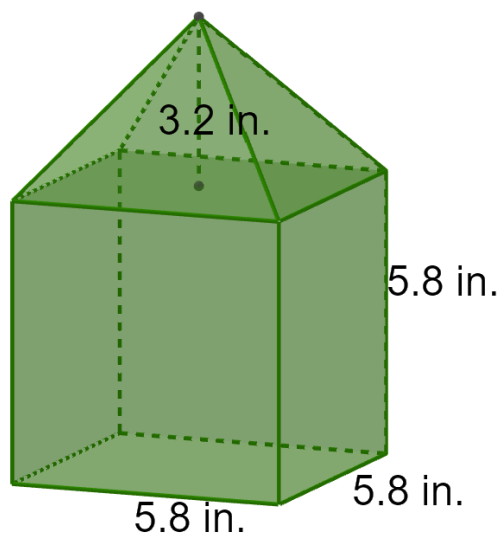




9.9.5 Wrap-Up: Ticket Out the Door

1. Jillian was building a birdhouse. Some of the dimensions, in inches (in.), of the birdhouse are shown. Determine the total amount of wood, in square inches, that Jillian will need to build the outside of the birdhouse.



Unit 9, Lesson 10: Determining How Dilations Affect Area



Benchmark

MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.



Learning Target

- ☐ I can discover the relationship between area and the scale factor of a dilation.



9.10.1 Warm-Up: Where Is Here?

A picture of a directional sign is shown.

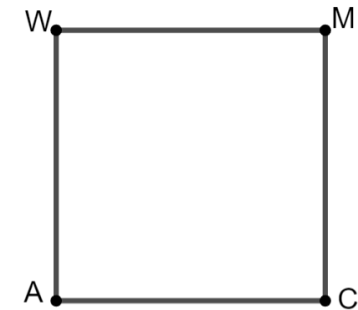


1. What do you notice about the sign?
2. Where do you think the picture was taken?
3. The Earth's average polar circumference is 24,859 miles. What do you think the sign means?



9.10.2 Exploration: Scale Factor and Area

1. Square $AWMC$ is shown, where $WM = 1$ unit.



- a. Determine the area of $AWMC$.
- b. $AWMC$ is dilated using point A as the center of dilation using each of the given scale factors. Complete the table.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation					
Area of Dilated Square					

- c. Write a conjecture about the relationship between the length of the scale factor and the value of the area.

- d. Determine the area of $AWMC$ if $WM = 4$ units.

2. Suppose square $DTSC$ is dilated using point D as the center of dilation for each of the given scale factors. Complete the table for $DTSC$ given that $TS = 4$ units.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation					
Area of Dilated Square					

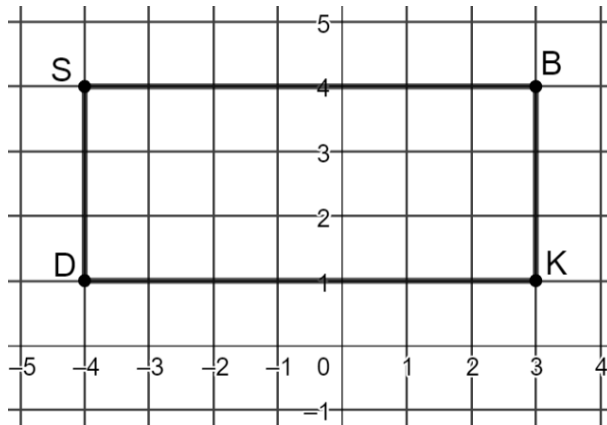
- a. Review your conjecture from the previous problem and make any revisions, if needed.

- b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

Summary of Partner's Conjecture	Partner's Initials

- c. Review your conjecture and make any revisions, if needed.

3. Rectangle $SBKD$ is shown.



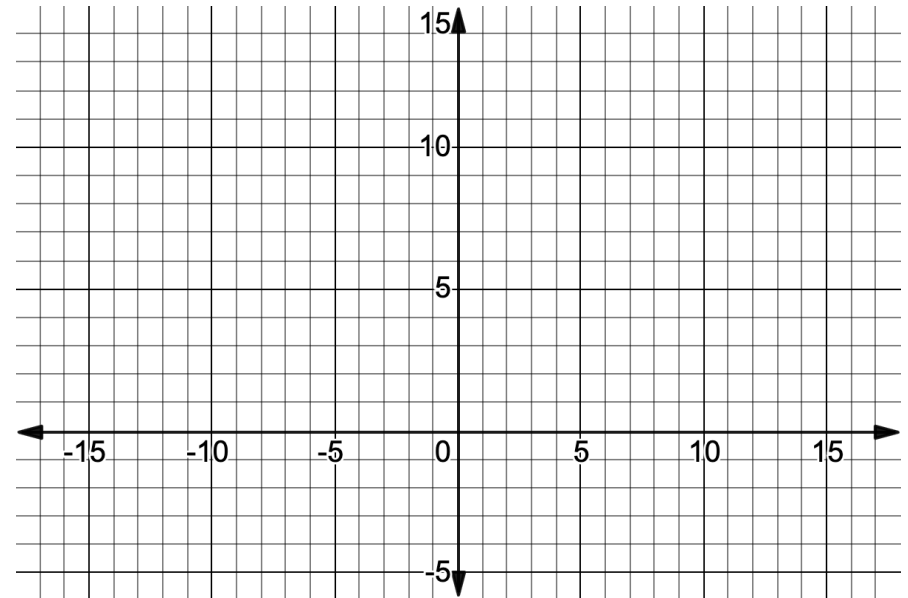
a. Using the coordinate grid, find the area of the rectangle.

Area=_____ square units

b. Dilate $SBKD$ by a scale factor of 3 centered at the origin. Write the new coordinates in the table.

Point	Preimage	Image
S	$(-4, 4)$	
B	$(3, 4)$	
K	$(3, 1)$	
D	$(-4, 1)$	

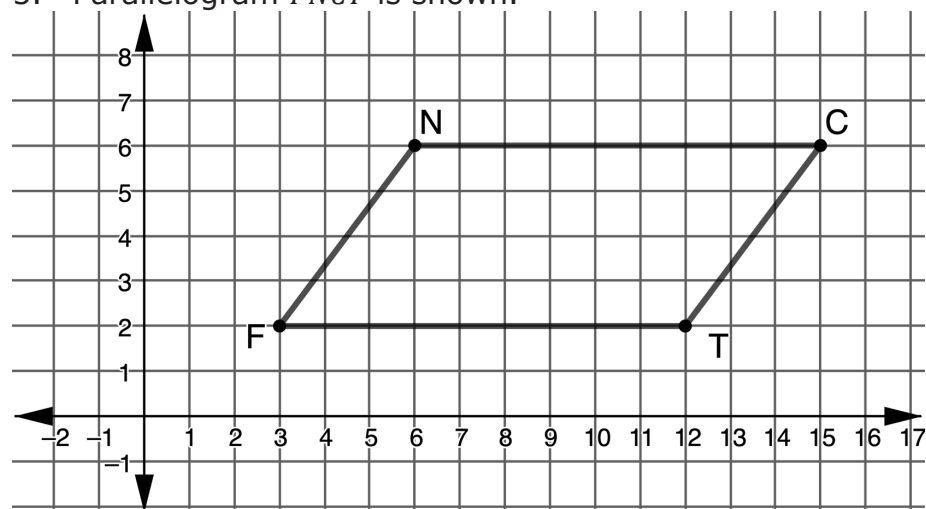
4. Graph $S'B'K'D'$ on the coordinate plane.



a. The area of rectangle $S'B'K'D'$ is _____ square units.

b. Does the conjecture you wrote about the relationship being area and scale factor when dilating a square hold true for rectangle $SBKD$?

5. Parallelogram $FNCT$ is shown.



a. What is the area of $FNCT$, in square units?

b. Dilate $FNCT$ by a scale factor of $\frac{2}{3}$ centered at the origin. Write the coordinates of the preimage and image in the table.

Point	Preimage	Image
F		
N		
C		
T		

c. Plot the coordinates of $F'N'C'T'$ on the same coordinate plane as $FNCT$.

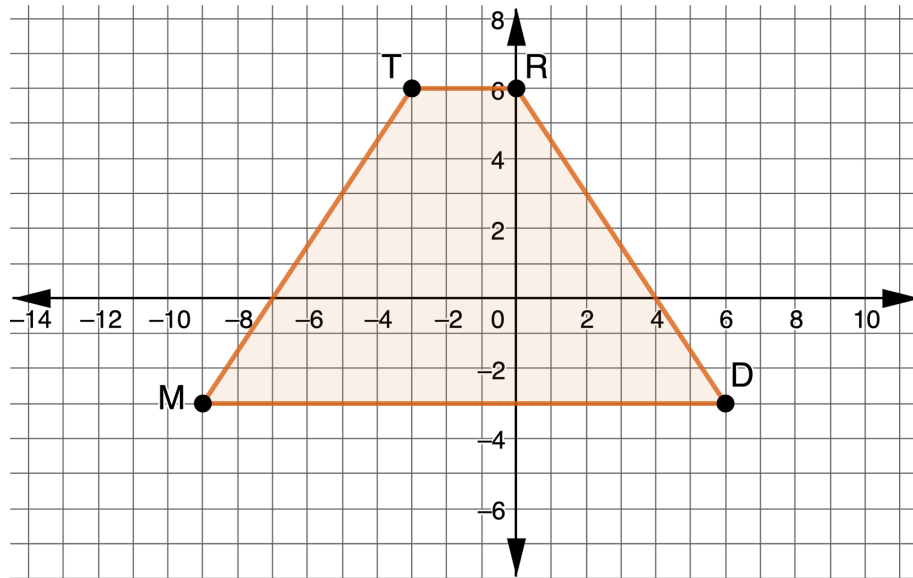
d. What is the relationship between the area of $FNCT$ and $F'N'C'T'$?

e. How can the scale factor of $\frac{2}{3}$ be used to determine the area of $F'N'C'T'$ from the area of $FNCT$?



9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area

1. Trapezoid $MTRD$ is shown on the coordinate plane.



- a. Find the area of $MTRD$.

$$A = \frac{1}{2}(\text{ } + \text{ })(\text{ })$$

The area of $MTRD$ is _____ square units.

- b. Find the coordinates of $MTRD$ given the scale factors in the table. Use the preimage coordinates to find the coordinates of the images. Both dilations are centered at $(0,0)$.

Point	Preimage	Dilation $k = \frac{1}{6}$ ($M'T'R'D'$)	Dilation $k = \frac{4}{3}$ ($M''T''R''D''$)
M			
T			
R			
D			

- c. Find the area of $M'T'R'D'$.

- d. What effects does the scale factor of $\frac{1}{6}$ have on the original area?

e. Find the area of $M''T''R''D''$.

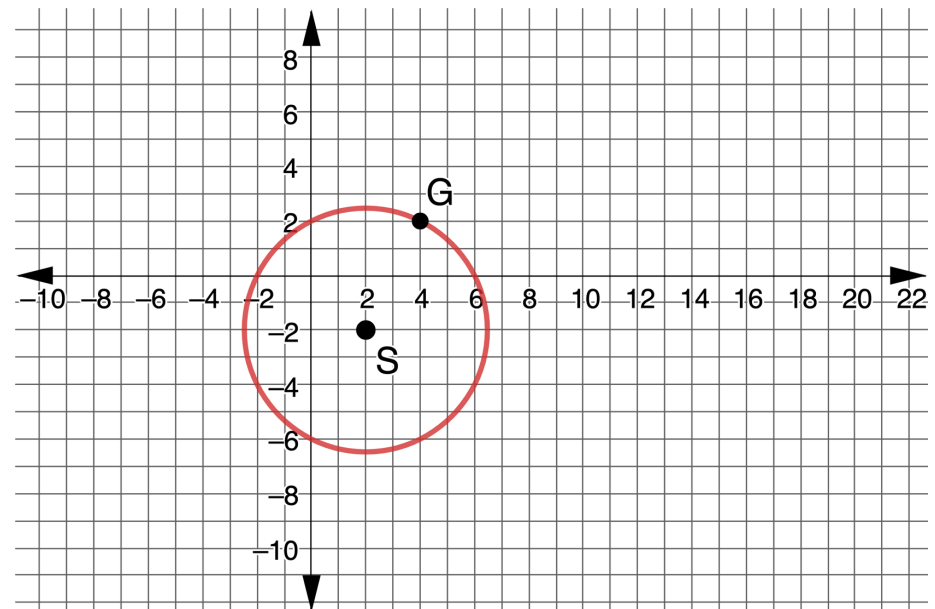
f. What effects does the scale factor of $\frac{4}{3}$ have on the original area?

g. How does the area of the trapezoid with a scale factor of $\frac{1}{6}$ differ from the area of the trapezoid with a scale factor of $\frac{4}{3}$?



9.10.4 Your Turn!

1. Circle S is shown with a radius of $\sqrt{20}$.



a. Dilate circle S using a scale factor of 1.5 centered at the origin, and graph the resulting image on the coordinate plane.

b. Find the area of the circles.

Area of Preimage	Area of Image



9.10.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about how dilations affect area.